

Auditing Project Report

Facial recognition technology is everywhere. I use it to unlock my phone and sort through photo libraries, while law enforcement uses it to identify suspects. But just because a system works well for some people does not mean it works well for everyone. As Ruha Benjamin argues in *Race After Technology*, automated systems can reinforce existing social inequalities even when they appear neutral (Benjamin, 2019). An algorithm trained mostly on one type of face may struggle with others, and this is why algorithmic audits matter. An audit takes a system, runs it against a known dataset, and checks whether the results are consistent across different groups. Brown describes this as a way to hold automated systems accountable by measuring whether their outputs are fair across demographic lines (Brown et al., 2021). Without audits, biased systems can make it into the real world and affect real people.

I decided to audit Amazon Rekognition, a facial recognition service that Amazon has sold to law enforcement and reportedly pitched to Immigration and Customs Enforcement (Fazzini, 2018). The American Civil Liberties Union (ACLU) has publicly criticized the technology, warning that "by automating mass surveillance, facial recognition systems like Rekognition threaten this freedom, posing a particular threat to communities already unjustly targeted in the current political climate" (Fazzini, 2018). Given that, I wanted to see for myself how well the system performs across different demographics. I tested its gender and age predictions on 1,000 face images from the FairFace dataset, which is balanced across race, gender, and age (Karkkainen and Joo, 2021).

I found that it does not treat everyone the same. Rekognition performed well overall, averaging around 9 out of 10 across most groups, but Black faces scored the lowest at 8.54, and Black males had the highest gender misidentification rate at 16.4%, which is more than triple the rate for some other groups. Before I break down what those numbers mean, let me show you my process.

Rekognition's `detect_faces` API takes an image and returns a predicted gender and an estimated age range. To automate the process, I wrote a Python script that sends each of the 1,000 FairFace images to the API, collects the predictions, and compares them against the validated labels. Each image in the dataset came with labels for age (e.g., 20-29), gender (Male or Female), and one of seven racial/ethnic groups: White, Black, Indian, East Asian, Southeast Asian, Middle Eastern, and Latino/Hispanic.

I originally started with Face++, but I kept hitting rate limits that cut me off partway through. After struggling with that for a while, I switched to Rekognition, which processed everything without interruption.

For quality ratings, every image starts at 10 points, and I subtract based on errors. Gender carries the largest penalty of -5 if the prediction is wrong, and I weighted it heavily because gender is either right or wrong, with no in-between. Age is more forgiving, though, because both come as ranges, not single numbers. But I did make an adjustment because

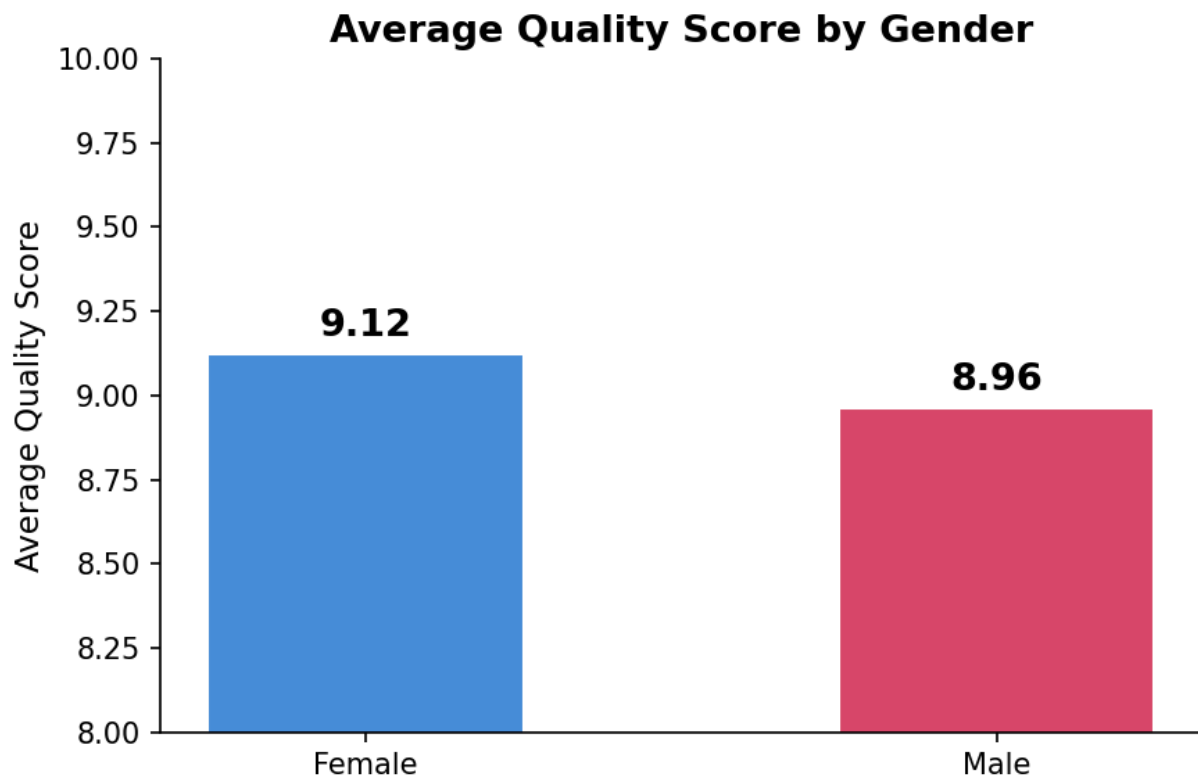
when I started reviewing the results, I noticed that Rekognition was suspiciously accurate, like almost everything scored highly. I realized it was because the age ranges barely overlapped by 1 year (e.g., predicted 10-20 vs. truth 20-29), and my script was counting that as a perfect match. That didn't feel fair, so I updated my scoring to count only the years that overlap between the two ranges. If the overlap is 5+ years, no deduction. If only 1-4 years overlap, it gets a -1 point deduction, and IMO this made the ratings feel much more honest and meaningful.

Rekognition does not predict race or ethnicity, so I could not score on that, but I still used the race/ethnicity labels from the dataset to compare average scores across groups, though, which is where the most interesting findings came from.

Across 1,000 images, Rekognition scored slightly higher on female faces (9.12) than male faces (8.96). The gap becomes clearer when you look at gender misidentification rates: 3.0% of female faces (13 out of 433) versus 6.7% of male faces (38 out of 567). Males were misidentified at more than double the rate. Since a wrong gender prediction costs 5 points, even a small increase in error rate drags down the average.

Table 1: Average Quality by Gender

Gender	Avg Score	Count	Misid Rate
Female	9.12	433	3.0%
Male	8.96	567	6.7%



The differences across racial groups were more pronounced. Middle Eastern faces scored the highest at 9.19, followed by Southeast Asian (9.14), Indian (9.14), East Asian (9.13), and White (9.10). Latino/Hispanic came in at 8.95. Black faces scored the lowest by a clear margin at 8.54.

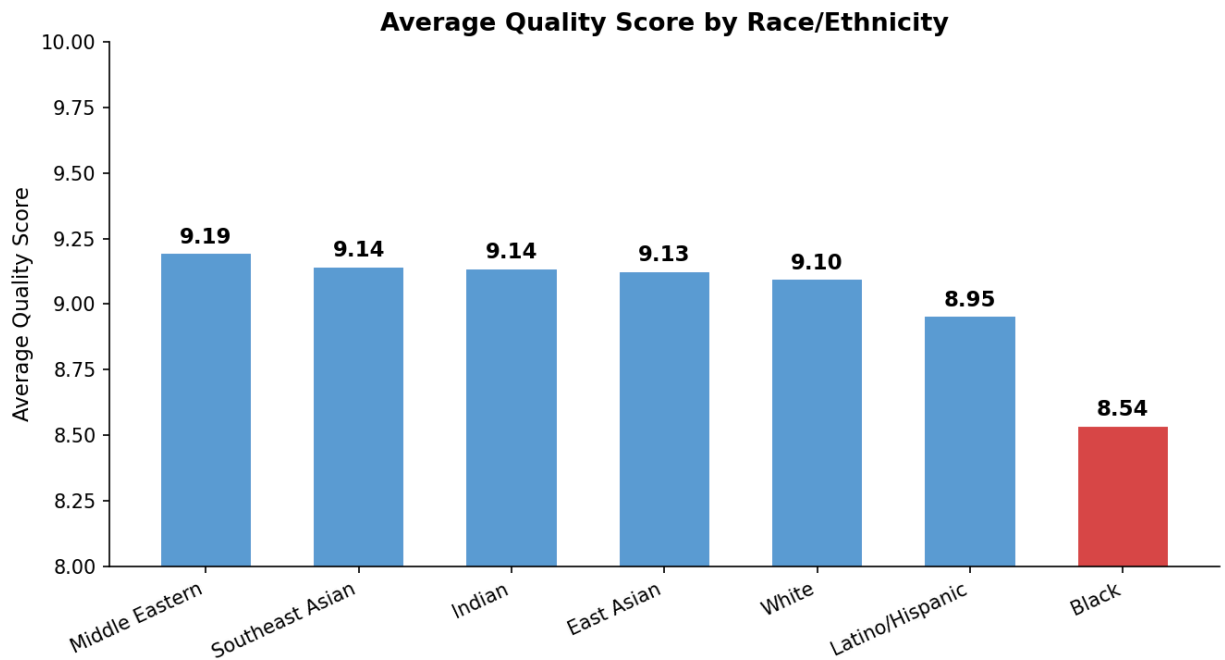
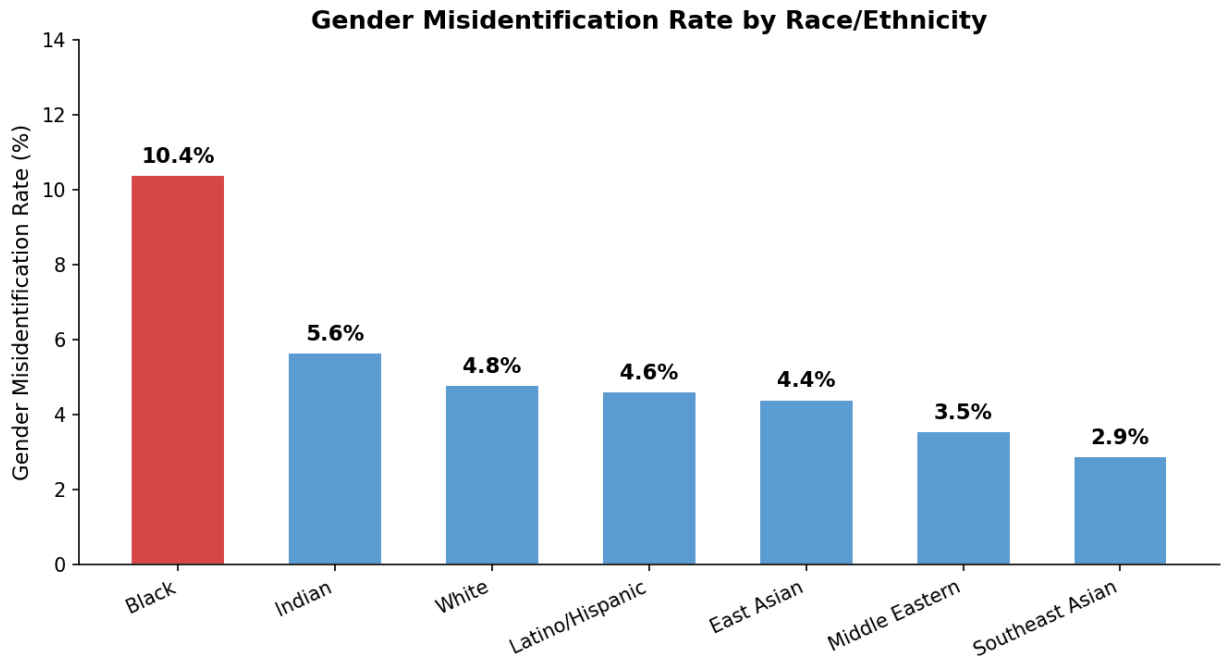


Table 2: Average Quality by Race/Ethnicity

Race/Ethnicity	Avg Score	Count	Misid Rate
Middle Eastern	9.19	113	3.5%
Southeast Asian	9.14	139	2.9%
Indian	9.14	124	5.6%
East Asian	9.13	159	4.4%
White	9.10	188	4.8%
Latino/Hispanic	8.95	152	4.6%
Black	8.54	125	10.4%

The reason becomes clear when you look at gender misidentification by race. Rekognition misidentified gender for 10.4% of Black faces, compared to 2.9% for Southeast Asian and 3.5% for Middle Eastern faces. That rate is roughly three times the average for other groups, and since gender errors carry a 5-point penalty, this one issue probably drives most of the score gap.



Breaking the scores down by both race and gender further shows that Black males averaged the lowest score of any group at 8.34, while Middle Eastern females scored the highest at 9.41.

Table 3: Average Quality by Race/Ethnicity and Gender

Race/Ethnicity	Female	Male
Middle Eastern	9.41	9.09
Southeast Asian	9.30	9.04
Indian	9.23	9.05
East Asian	9.16	9.09
Latino/Hispanic	9.12	8.79
White	9.00	9.16
Black	8.76	8.34

I found that there is a measurable bias in Rekognition's performance against Black faces, and that the software performs well overall, but worst on the group that already faces the most

discrimination in automated systems (Benjamin, 2019). The bias is not dramatic enough to make Rekognition useless, but it is important to recognize this flaw.

I feel like the most likely explanation is training data. Karkkainen and Joo showed that most major face datasets are heavily skewed toward White faces, with other groups severely underrepresented. If Rekognition was built on similarly imbalanced data, the model simply had less practice with Black faces.

I remember the readings describing this as the "New Jim Code," in which technology reproduces racial inequality while appearing objective (Benjamin, 2019). Benjamin argues these errors are not random glitches but signals of a deeper problem. When Rekognition misidentifies a Black male's gender at 16.4% while getting it wrong for Southeast Asian faces only 2.9% of the time, that is a pattern, and it reflects whose faces were prioritized when the system was built. As she writes, "technical systems offer a mirror to the wider terrain of struggle over the forces that govern our lives" (Benjamin, 2019).

Brown argues that audits need to connect their findings to the people affected (Brown et al., 2021). Isn't it crazy that this same software was pitched to ICE for immigration enforcement, so real people's lives are getting affected by systems with measurable racial bias (Fazzini, 2018). Although this audit might not fix the problem, it can make it visible.

Fleming and Bruce write that "the absence of harmful intentions alone is not enough to prevent models from producing harmful predictions" (Fleming and Bruce, 2021). I'm sure the engineers who built Rekognition did not intend to create a biased system, but if no one checked whether the training data adequately represented Black faces before deployment, that unfortunately led to one.

I want to draw a comparison to when police records in Puerto Rico were missing up to 27% of femicide victims, and this was because the system was not designed to see them (D'Ignazio, 2020). I feel like this same logic holds for facial recognition, because underrepresentation in training data directly translates into lower accuracy.

The class readings prepared me for the bias that results from building systems on incomplete data without sufficient testing. Although this audit might not fix the problem, it can make it visible.

Works Cited

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